

LONDON PETRO ACADEMY
Empowering Professionals for a Sustainable Future



Drilling for Non-Drilling Personnel

A Practical Introduction to Oil & Gas Well Planning, Rig Operations, and Drilling Costs

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Drilling for Non-Drilling Personnel

This 3 Day course provides a comprehensive overview of the fundamentals of oil and gas drilling engineering concepts and day-to-day operations.

It elaborates on drilling terminologies commonly used in drilling operations, rig equipment and their functions. If you're new to drilling and looking for a comprehensive overview of how it all really works. This brand-new “3 Days: Drilling for Non-Drilling Personnel” course is designed with you in mind. This hands-on training course will walk you through drilling operations, technology and costs and what it takes to plan and drill a well, as well as the logistics and roles of various service companies involved in the process. At the end of the course, you will be literate in oil and gas and be able to read and understand a daily drilling report!

Fundamentals of drilling operations; the drilling rig equipment and their functions; drilling terminologies; drilling processes; planning of drilling operations; logistics, role and responsibilities of service companies including drilling contractor; the functions of drilling fluid, casing, cementing and components of a drill string; the different types of drilling contracts and the types of wells; and, HSE and resources requirements during drilling.



LOCATION

London

Houston

Dubai

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COURSE OBJECTIVES

By the end of this 3-day course, participants will be able to:

- 1.Explain how oil and gas are formed, trapped in the subsurface, and discovered through exploration activities.
- 2.Describe the main drilling rig types (land and offshore) and identify the key rig systems and their functions (power, hoisting, circulating, rotating, well control).
- 3.Define common drilling terminology and outline the roles and responsibilities of the operating company, drilling contractor, and major service companies.
- 4.Summarise the well planning process, including data requirements, basic well design, directional drilling concepts, and preparation of an AFE and well cost estimate.
- 5.Describe the functions and basic properties of drilling fluids, drill string components, casing, and cementing, and recognize typical drilling problems and non-productive time drivers.
- 6.Outline the principles of well control, including kicks, blowout prevention equipment, shut-in and well kill concepts.
- 7.Recognise key logistics requirements and support services for drilling operations, including rig location, marine / aviation support, and supply base needs
- 8.Identify major health, safety, environment and security (HSES) risks in drilling operations and the purpose of PPE, hazard control measures, and emergency response.
- 9.Describe the basic concepts of well completion, testing, artificial lift and workover operations and how they interface with drilling.
- 10.Read and interpret the main sections of a daily drilling report and relate the reported data to actual rig activities, costs, and performance



This course is ideal for non-drilling professionals who interact with drilling projects and need a solid understanding of drilling operations, costs, risks and terminology.

It is particularly suitable for:

- Reservoir engineers and petrophysicists
- Geoscientists and subsurface professionals
- Economists, planners and facilities planning engineers
- Accountants, finance and mid-level management involved in E&P projects
- Personnel from drilling and oilfield service companies
- Logistics, procurement, IT, HR and HSE staff supporting drilling operations
- Executive assistants, secretaries and other technical support staff in drilling-related departments

Day 1

Introduction to Oil & Gas Industry

Oil and Gas Development

Formations, rock pressure, and hydrocarbon traps

Petroleum reservoirs

Exploration processes

Introduction to Drilling

Definitions and terminologies

- Drilling rig team
- Oil company structure
- Drilling contractor responsibilities
- Drilling service companies responsibilities

Drilling rig types

- Land rigs
- Offshore rigs

Drilling Rig Components

Drilling rig components and functions

- Rig power system
- Hoisting system
- Circulating system

Rotating system

- Drilling data recording system

Well control system

Well Planning and Design

Well planning and design process

- Data inputs
- Long lead time items and procurement
- Offset wells reviews
- Well Cost Estimate (AFE)
- Drilling fluids design
- Well trajectories
- Directional drilling



Day 2

Drilling Bits

Types of bits

- Classification of bits
- Rock failure mechanisms
- Bit records
- Components of drill string

Drilling Operations and Hole Cleaning

- Well drilling activities
- Drilling parameters
- Drilled cuttings handling
- Drilling fluids (mud)
- Typical drilling problems and lost time

Functions of Drilling Fluids

Mud properties

Water-based muds

Oil-based muds

Hydrostatic pressure

Buoyancy

Rheological models

Casing and Cementing

Functions of casing and cementing

Properties of cement

Single stage and multi-stage cementing

Blowout Prevention System

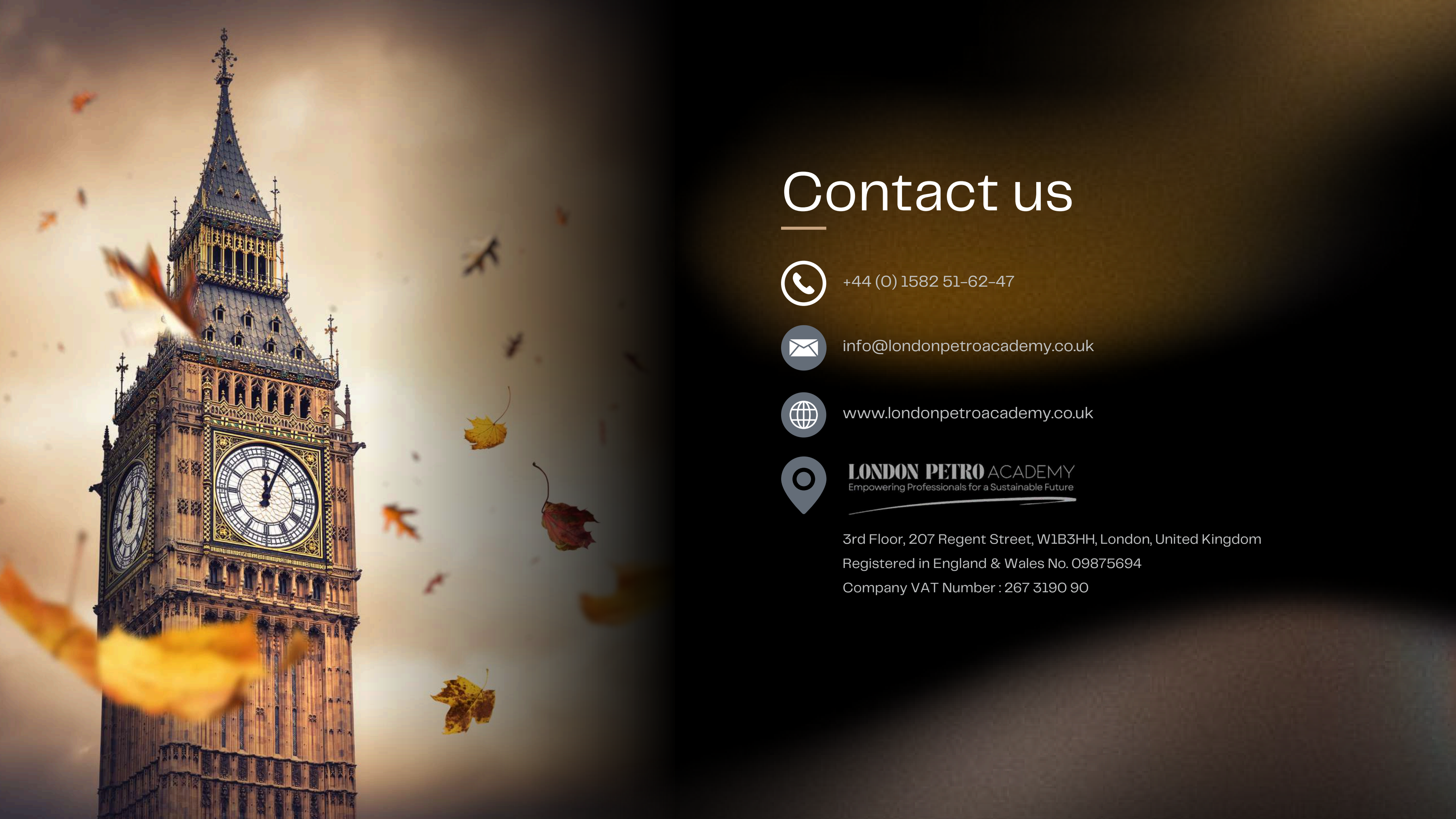
- Well control
- Kick and its causes
- Prevention of kicks
- Well Shut-in
- Well kill operations



Day 3

- Logistic Support and Services
- Rig location and access to services
- Weather conditions
- Supply vessels
- Helicopters
- Land base
- • Sea port facility
- Health, Safety, Environment and Security
- People and safety
- Personal Protective Equipment (PPE)
- Drilling operations and equipment safety
- Slip and trip
- Fall protection
- Hazardous Energy
- Chemical Hazards
- Fire Safety
- Hydrogen sulfide safety
- Oil spill prevention and response
- First Aid
- Well Completion
- Types of completions
- Perforating a well
- Well testing
- Completion equipment, concepts and techniques
- Multizone completions
- Artificial lift technique
- Workover operations
- Wireline pressure control equipment and Christmas tree





Contact us



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